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Sensor device package and image display device including the same

Abstract

A sensor device package includes a sensor device including touch sensing electrodes and an antenna unit, and a circuit board bonded to the sensor device. The circuit board includes a core layer including a first surface and a second surface which face each other, a first conductive layer including first touch sensor signal lines and a first antenna signal line distributed at the same level on the first surface, a second conductive layer including second touch sensor signal lines and a second antenna signal line distributed at the same level on the second surface, an antenna via structure which penetrates the core layer to connect the first antenna signal line and the second antenna signal line with each other, and a touch sensor via structure which penetrates the core layer to connect the first touch sensor signal lines and the second touch sensor signal lines to each other, respectively.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION AND CLAIM OF PRIORITY
(1) This application claims the benefit under 35 USC § 119 of Korean Patent Application No. 10-2023-0143425 filed on Oct. 25, 2023, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference for all purposes.
BACKGROUND
1. Field of the Invention
(2) The present invention relates to a sensor device package and an image display device including the same, and more specifically, to a sensor device package which includes a sensor device and a

circuit board, and an image display device including the sensor device package.

2. Description of the Related Art

(3) Recently, according to development of the information-oriented society, wireless communication techniques such as Wi-Fi, Bluetooth, and the like are implemented, for example, in a form of smartphones by combining with image display devices. In this case, an antenna may be coupled to the image display device to perform a communication function.

(4) Recently, with mobile communication techniques becoming more advanced, an antenna for performing high frequency or ultra-high frequency communication corresponding to 3G to 5G or higher, for example, may be coupled to the image display device.

(5) Meanwhile, electronic devices, in which a touch panel or a touch sensor as an input device for allowing a user to select instructions displayed on a screen by his or her finger or an object such as a touch pen and input his or her command is coupled with the image display device to implement an image display function and an information input function together, have been developed. For example, as disclosed in Korean Patent Laid-Open Publication No. 2014-0092366, a touch screen panel, in which a touch sensor is coupled to various image display devices, has been developed.

(6) When the antenna structure is disposed together with the touch sensor, signal interference between the antenna and the touch sensing electrode may occur. In addition, mutual signal interference may also occur in a circuit structure for performing power supply to the antenna structure and the touch sensor, respectively.

(7) For example, Korean Patent Laid-Open Publication No. 2003-0095557 discloses an antenna structure built into a portable terminal, but it fails to consider consistency with other electrical devices such as a touch sensor.

SUMMARY

(8) An object of the present invention is to provide a sensor device package having improved reliability and efficiency of signal transmission and reception.

(9) Another object of the present invention is to provide an image display device including the sensor device package having improved reliability and efficiency of signal transmission and reception.

(10) To achieve the above objects, the following technical solutions are adopted in the present invention. 1. A sensor device package including: a sensor device including touch sensing electrodes and an antenna unit; and a circuit board which is bonded to the sensor device, wherein the circuit board includes: a core layer including a first surface and a second surface which face each other; a first conductive layer including first touch sensor signal lines and a first antenna signal line which are distributed at the same level on the first surface of the core layer; a second conductive layer including second touch sensor signal lines and a second antenna signal line which are distributed at the same level on the second surface of the core layer; an antenna via structure which penetrates the core layer to connect the first antenna signal line and the second antenna signal line with each other; and a touch sensor via structure which penetrates the core layer to connect each of the first touch sensor signal lines and the second touch sensor signal lines with each other. 2. The sensor device package according to the above 1, wherein the sensor device further includes touch sensor pads electrically connected with the touch sensing electrodes and an antenna pad electrically connected with the antenna unit, wherein the first touch sensor signal lines of the circuit board are bonded to the touch sensor pads respectively, and the first antenna signal line of the circuit board is bonded to the antenna pad. 3. The sensor device package according to the above 2, wherein the touch sensor pads and the antenna pad are arranged at one end of the sensor device to form a single pad row. 4. The sensor device package according to the above 3, wherein the sensor device further includes a blocking pad which is included in the single pad row and inserted between adjacent touch sensor pads of the touch sensor pads, or a guard pad disposed at an end of the single pad row. 5. The sensor device package according to the above 4, wherein the sensor device further includes: traces which connect the touch sensing electrodes and the touch sensor pads with each other; and a

blocking line which extends from the blocking pad between adjacent traces of the traces. 6. The sensor device package according to the above 4, wherein the sensor device further includes a guard line of a loop shape, which extends from the guard pad and surrounds the touch sensing electrodes. 7. The sensor device package according to the above 3, further including one anisotropic conductive film which bonds the single pad row and the first conductive layer of the circuit board. 8. The sensor device package according to the above 2, wherein the antenna unit includes a radiator and a transmission line which extends from the radiator to be connected to the antenna pad. 9. The sensor device package according to the above 8, wherein a plurality of antenna units are arranged in a row direction, and the antenna pads are independently connected to each of the antenna units. 10. The sensor device package according to the above 9, wherein the transmission lines connected to each of the plurality of antenna units have a bent line shape and have the same length. 11. The sensor device package according to the above 9, wherein at least one touch sensor pad of the touch sensor pads is arranged between the antenna pads connected to adjacent antenna units of the antenna units. 12. The sensor device package according to the above 1, further including: a touch sensor driving circuit/touch sensor connection structure electrically connected with the second touch sensor signal lines on the second surface of the core layer; and an antenna driving circuit/antenna connection structure electrically connected with the second antenna signal line on the second surface of the core layer. 13. The sensor device package according to the above 12, wherein the antenna driving circuit/antenna connection structure includes an antenna driving integrated circuit chip or an antenna connector, and the touch sensor driving circuit/touch sensor connection structure includes a touch sensor driving integrated circuit chip or a touch sensor connector. 14. The sensor device package according to the above 1, wherein the circuit board further includes a mid-ground layer disposed between the first conductive layer and the second conductive layer in the core layer. 15. The sensor device package according to the above 14, wherein the antenna via structure and the touch sensor via structure are electrically separated from the mid-ground layer and penetrate the mid-ground layer. 16. The sensor device package according to the above 1, wherein the sensor device further includes a substrate layer, wherein the touch sensing electrodes and the antenna unit are arranged together at the same level on the upper surface of the substrate layer. 17. The sensor device package according to the above 16, wherein the substrate layer includes an active area where the touch sensing electrodes are distributed and a peripheral area surrounding the active area, wherein the antenna unit is partially disposed on the active area.

(11) According to embodiments of the present invention, the radiator and the sensing electrodes may be disposed together in the active area of the sensor device, thereby improving space efficiency. In addition, the touch sensor pads and the antenna pads are disposed together in the bonding area, such that power supply/signal transmission may be performed through one circuit board.

(12) In exemplary embodiments, the circuit board has a multilayer structure including a first conductive layer and a second conductive layer, and each of the first conductive layer and the second conductive layer may include an antenna signal line and a touch signal line. Through the antenna via structure and the touch via structure which connect the first conductive layer and the second conductive layer with each other, power supply reliability and power supply efficiency to the antenna pad and the touch sensor pad may be improved through the single circuit board.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other objects, features and other advantages of the present invention will be more clearly understood from the following detailed description taken in conjunction with the

accompanying drawings, in which:
(2) FIG. **1** is a schematic plan view illustrating a sensor device according to exemplary embodiments;
(3) FIG. **2** is a schematic cross-sectional view illustrating a touch sensing electrode structure of the sensor device according to exemplary embodiments;
(4) FIGS. **3** and **4** are schematic cross-sectional views illustrating a sensor device package according to exemplary embodiments;
(5) FIG. **5** is a schematic partially enlarged plan view illustrating the sensor device package according to exemplary embodiments;
(6) FIGS. **6** and **7** are schematic cross-sectional views illustrating the sensor device package according to exemplary embodiments; and
(7) FIGS. **8** and **9** are a schematic plan view and a cross-sectional view illustrating an image display device according to exemplary embodiments, respectively.
DETAILED DESCRIPTION
(8) Embodiments of the present invention provide a sensor device package which includes a sensor device including an antenna unit and sensing electrodes, and a circuit board. In addition, an image display device including the sensor device package is provided.
(9) The antenna unit may be, for example, a microstrip patch antenna manufactured in the form of a transparent film. The sensor device including the antenna unit may be applied to, for example, a communication device for high frequency or ultra-high frequency (e.g., 3G, 4G, 5G or higher) mobile communication. However, the use of the sensor device is not limited only to the image display device, and the sensor device may be applied to various structures such as a vehicle, a home appliance, a building and the like.
(10) Hereinafter, preferred embodiments of the present invention will be described in detail with reference to the accompanying drawings. However, since the drawings attached to the present disclosure are only given for illustrating one of several preferred embodiments of present invention to easily understand the technical spirit of the present invention with the above-described invention, it should not be construed as limited to such a description illustrated in the drawings.
(11) The terms “first,” “second,” “one surface,” “the other surface,” “one end,” “the other end,” “upper side,” “lower side,” “upper portion,” “lower portion,” “column direction,” “row direction,” and the like as used herein do not limit the absolute position or order, but are used in a relative sense to distinguish different components or portions.
(12) The term “row direction” as used herein may correspond to a width direction parallel to an active surface or display surface of a sensor device or an image display device. The term “column direction” may be a length direction parallel to the active surface or the display surface, and may be a direction perpendicular to the row direction.
(13) FIG. **1** is a schematic plan view illustrating a sensor device according to exemplary embodiments. FIG. **2** is a schematic cross-sectional view illustrating a touch sensing electrode structure of the sensor device according to exemplary embodiments. FIG. **2** is a cross-sectional view taken on line I-I′ of FIG. **1** in a thickness direction.
(14) Referring to FIGS. **1** and **2**, the sensor device may include a substrate layer **100**, touch sensing electrodes **110** and **130** (hereinafter, abbreviated as sensing electrodes) and an antenna unit **200** which are disposed on the substrate layer **100**.
(15) The substrate layer **100** may include a support layer, an interlayer insulation layer or a film type substrate for forming the sensing electrodes **110** and **130** and the antenna unit **200**. For example, the substrate layer **100** may also be provided as a dielectric layer of the antenna unit **200**.
(16) For example, the substrate layer **100** may include a transparent resin film including a polyester resin such as polyethylene terephthalate, polyethylene isophthalate, polyethylene naphthalate, polybutylene terephthalate, etc.; a cellulose resin such as diacetyl cellulose, triacetyl cellulose, etc.; a polycarbonate resin; an acrylic resin such as polymethyl (meth)acrylate, polyethyl (meth)acrylate,

etc.; a styrene resin such as polystyrene, acrylonitrile-styrene copolymer, etc.; a polyolefin resin such as polyethylene, polypropylene, cyclic polyolefin or polyolefin having a norbornene structure, ethylene-propylene copolymer, etc.; a vinyl chloride resin; an amide resin such as nylon, aromatic polyamide; an imide resin; a polyether sulfonic resin; a sulfonic resin; a polyether ether ketone resin; a polyphenylene sulfide resin; a vinylalcohol resin; a vinylidene chloride resin; a vinylbutyral resin; an allylate resin; a polyoxymethylene resin; an epoxy resin; a urethane or acrylic urethane resin, a silicone resin and the like. These may be used alone or in combination of two or more thereof.

(17) In some embodiments, the substrate layer **100** may include an adhesive film, such as an optically clear adhesive (OCA), an optically clear resin (OCR) and the like.

(18) In some embodiments, the substrate layer **100** may include an inorganic insulation material such as a silicon oxide, silicon nitride, silicon oxynitride, glass and the like.

(19) In one embodiment, the substrate layer **100** may be provided as a substantial single layer.

(20) In one embodiment, the substrate layers **100** may each include a multilayer structure of two or more layers. For example, the substrate layer **100** may include a lower substrate and a dielectric layer, and may include an adhesive layer between the lower substrate and the dielectric layer.

(21) An impedance or inductance for the antenna unit **200** is formed by the substrate layer **100**, thus to adjust a frequency band which can be driven or sensed by the antenna unit **200**. In some embodiments, the dielectric constant of the substrate layer **100** may be adjusted to a range of about 1.5 to 12. When the dielectric constant exceeds about 12, a driving frequency is excessively reduced, such that driving (of the antenna) in a high frequency band may not be implemented.

(22) In one embodiment, a ground layer (not shown) may be disposed under a lower surface of the substrate layer **100**.

(23) In one embodiment, a conductive member of an image display device or display panel to which the sensor device is applied may be provided as the ground layer.

(24) For example, the conductive member may include electrodes or wirings such as a gate electrode, source/drain electrodes, a pixel electrode, a common electrode, a data line, a scan line, and the like included in a thin film transistor (TFT) array panel.

(25) In one embodiment, a metallic member such as a stainless steel (SUS) plate, a sensor member such as a digitizer, a heat radiation sheet, etc., disposed on a back portion of the image display device may be provided as the ground layer.

(26) The substrate layer **100** may include, on an upper surface thereof, an active area AA and a peripheral area PA around the active area AA. The sensing electrodes **110** and **130** and the antenna unit **200** may be arranged on the active area AA of the substrate layer **100**.

(27) The sensing electrodes **110** and **130** may be disposed on the upper surface of the substrate layer **100** in the active area AA portion. When a touch is input form a user into the active area AA, a change in electrostatic capacity may occur by the sensing electrodes **110** and **130**. Accordingly, a physical touch may be converted into an electrical signal to perform a predetermined sensing function.

(28) The sensing electrodes **110** and **130** may include first sensing electrodes **110** and second sensing electrodes **130**. The first sensing electrodes **110** and the second sensing electrodes **130** may be arranged in a direction intersecting each other. The first sensing electrodes **110** and the second sensing electrodes **130** may be positioned in the same layer or at the same level as each other on the upper surface of the substrate layer **100**.

(29) For example, the first sensing electrodes **110** may be arranged in a column direction (e.g., a Y direction). The first sensing electrodes **110** may be connected with each other in the column direction through a mergence part **115**. The mergence part **115** may be integrally connected with the first sensing electrodes **110** to be provided as a substantial single member.

(30) When a plurality of first sensing electrodes **110** are connected by the mergence parts **115**, sensing channel columns extending in the column direction may be defined. In addition, a plurality

of the sensing channel columns may be arranged in a row direction (e.g., an X direction).
(31) The second sensing electrodes **130** may be arranged in the row direction. Each of the second sensing electrodes **130** may have an island pattern shape spaced apart from each other. The second sensing electrodes **130** adjacent to each other in the row direction may be electrically connected with each other by a bridge electrode **135**.
(32) For example, a pair of second sensing electrodes **130**, which are adjacent to each other with the mergence part **115** included in the sensing channel columns interposed therebetween, may be electrically connected with each other by the bridge electrode **135**. Accordingly, a plurality of sensing channel rows may be defined by the second sensing electrodes **130** and the bridge electrodes **135** connected in the row direction. In addition, the plurality of the sensing channel rows may be arranged in the column direction.
(33) As shown in FIG. **2**, an interlayer insulation layer **120** which covers the first and second sensing electrodes **110** and **130** is formed, and the bridge electrode **135** may penetrate the interlayer insulation layer **120** to connect the adjacent second sensing electrodes **130**.
(34) In some embodiments, a protective layer **190** which covers the bridge electrode **135** may be formed on the interlayer insulation layer **120**. The interlayer insulation layer **120** and the protective layer **190** include the above-described resin material or inorganic insulation material, and may cover the sensing electrodes **110** and **130** and the antenna units **200** together. The interlayer insulation layer **120** and the protective layer **190** may be removed in a bonding area BA, such that the pads are exposed to an outside.
(35) In FIG. **1**, it is illustrated that each of the sensing electrodes **110** and **130** has a rhombus pattern shape, but the shapes of the sensing electrodes **110** and **130** may be appropriately changed in consideration of the pattern density, consistency with optical characteristics of the image display device, arrangement of the antenna units **200** and the like. For example, the sensing electrodes **110** and **130** may be formed to have a wavy edge.
(36) In FIG. **1**, it is illustrated that column-directional sensing electrodes are integrally connected with each other by the mergence parts, and row-directional sensing electrodes are connected with each other by the bridge electrodes, however, the column direction and the row direction are used relatively to refer to two other intersecting directions, and it is not intended to limit them to a specific direction.
(37) In addition, the number of sensing channel rows and sensing channel columns shown in FIG. **1**, and the number of sensing electrodes included therein are only partially shown for the convenience of description, and may be expanded according to an area of the active area AA.
(38) An area of the upper surface of the substrate layer **100** except for the active area AA may be defined as a peripheral area PA. The peripheral area PA may be defined as an area which at least partially surrounds the periphery of the active area AA. The peripheral area PA may include a bezel area of the image display device, and may include the bonding area BA in which the pads are arranged. The active area AA may be overlapped with a display area of the image display device.
(39) The sensor device may further include traces **140** and touch sensor pads **150**.
(40) The traces **140** may be branched from each sensing channel row and sensing channel column to extend on the peripheral area PA. The traces **140** may include first traces **142** which are branched from the sensing channel columns and second traces **144** which are branched from the sensing channel rows. The traces **140** may include a conductive material and/or lamination structure substantially the same as or similar to the sensing electrodes **110** and **130**.
(41) In some embodiments, the second traces **144** may be arranged in accordance with a double-routing manner. The second traces **144** may be alternately distributed on both sides of the peripheral area PA in the row direction.
(42) For example, the second traces **144** may be arranged alternately in the column direction on the both sides. A second trace **144** may be branched from an end on one side of a sensing channel row among the sensing channel rows, and another second trace **144** may be branched from an end on a

side opposite the one side of another sensing channel row adjacent to the sensing channel row.
(43) By the above-described double-routing arrangement, a sufficient active area AA area may be secured while reducing the area of the peripheral area PA on both sides. In addition, by reducing a deviation in the length of the second traces **144**, resistance/sensing uniformities may be improved. Further, by distributing the conductive lines on both sides, a stress due to folding of the sensor device may be uniformly dispersed.
(44) In some embodiments, the first traces **142** may be branched from one end of each of the sensing channel columns adjacent to the bonding area BA. Accordingly, the length of the first traces **142** may be decreased, thereby increasing sensing/signal transmission speeds.
(45) In one embodiment, some of the first traces **142** may be branched from the other end of the sensing channel column and extend to the bonding area BA.
(46) As shown in FIGS. **1** and **2**, the sensing electrodes **110** and **130** may be arranged in accordance with mutual capacitance manner.
(47) In some embodiments, the sensing electrodes **110** and **130** may also be arranged in accordance with a self-capacitance manner. In this case, each of the sensing electrodes **110** and **130** may have an independent island pattern shape, and the traces **140** may be branched from each of sensing electrodes **110** and **130** having the island pattern shape. In addition, the mergence part **115** and the bridge electrode **135** may be omitted.
(48) The traces **140** may extend on the peripheral area PA and may be collected in the bonding area BA. Touch sensor pads **150** which are connected to distal ends of the traces **140** may be disposed within the bonding area BA.
(49) The touch sensor pads **150** may include first touch sensor pads **152** electrically connected with each of the sensing channel columns through the first traces **142** and second touch sensor pads **154** electrically connected with each of the sensing channel rows through the second traces **144**.
(50) In some embodiments, a blocking pad **151** and a guard pad **153** may be arranged adjacent to the touch sensor pads **150**. The blocking pad **151** and the guard pad **153** may not be electrically or physically connected with the traces **140**. The blocking pad **151** and the guard pad **153** are not connected with the touch sensing electrodes **110** and **130**, and the first and second touch sensor pads **152** and **154**, and are arranged independently, but may be included in a configuration which enhances the reliability of touch sensing/driving.
(51) The blocking pad **151** may be disposed between the first touch sensor pad **152** and the second touch sensor pad **154** adjacent to each other. For example, a plurality of first touch sensor pads **152** may be arranged in the row direction to form a first touch pad row, and a plurality of second touch sensor pads **154** may be arranged in the row direction to form a second touch pad row. The blocking pad **151** may be inserted between the first touch pad row and the second touch pad row.
(52) The independence of current and signal in the sensing channel row and the sensing channel column is enhanced by the blocking pad **151**, and mutual interference between a drive current and a receiving current may be prevented.
(53) The guard pad **153** may be disposed at an end of the pad row arranged in the bonding area BA. For example, the guard pad **153** may be the outermost pad included in the pad row.
(54) In some embodiments, the guard pads **153** may be disposed at both ends of the pad row.
(55) According to exemplary embodiments, a blocking line **141** and a guard line **143** may be connected to blocking pads **151** and the guard pads **153**, respectively. The blocking line **141** and the guard line **143** may extend from the blocking pads **151** and the guard pads **153**, respectively, within the peripheral area PA.
(56) As shown in FIG. **1**, an area where the first traces **142** extend in the row direction and an area where second traces **144** extend in the row direction may be divided by the blocking line **141**. Due to the blocking line **141**, signal collisions between the first traces **142** and the second traces **144** may be prevented, and mutual interference between the drive current and the receiving current may be prevented.

(57) The guard line **143** may extend to at least partially surround the active area AA. For example, the guard line **143** may extend continuously between the guard pads **153** disposed at both ends of the pad row to form a loop. Accordingly, mutual interference between a touch signal and an external noise in the active area AA may be reduced/blocked overall.
(58) The antenna unit **200** may include a radiator **210** ground lines **230** a transmission line **220**.
(59) For example, the radiator **210** may have a polygonal plate shape.
(60) For example, the transmission line **220** may have a width smaller than that of the radiator **210** and may be connected with one end or one side of the radiator **210**. The radiator **210** and the transmission line **220** may be formed as a single member integrally connected with each other.
(61) A target resonance frequency of the antenna unit **200** may be adjusted depending on the shape/size of the radiator **210**. For example, the radiator **210** may be designed to radiate radio waves in a high frequency/ultra-high frequency band of 3G, 4G, 5G or higher. For example, a radiation band of a frequency band of about 0.5 GHz or higher, about 1 GHz or higher, about 10 GHz or higher, about 20 GHz or higher, about 30 GHz or higher, or about 40 GHz or higher may be implemented through the radiator **210**.
(62) Antenna pads **250** may be arranged together with the touch sensor pads **150** within the bonding area BA. Antenna power supply/driving signals for the antenna units **200** may be transmitted from the antenna pads **250** to the transmission lines **220**.
(63) In some embodiments, the transmission line **220** may have a variable width. As illustrated in FIG. **1**, the transmission line **220** may include an extended line part **225** whose width is increased at an end adjacent to the radiator **210**. The extension line part **225** may suppress signal loss transmitted from the antenna pad **250** to the radiator **210** and facilitate impedance matching with the radiator **210**.
(64) The transmission lines **220** have a bent line shape, respectively, thereby allowing the antenna pads **250** to be efficiently collected within the bonding area BA. For example, the transmission lines **220** may include two bent portions, respectively.
(65) In some embodiments, all the lengths of the transmission lines **220** of the antenna units **200** may be substantially the same. In this case, it is possible to make signal phases applied to the respective antenna units **200** be uniform. Accordingly, radiation of a desired frequency band may be implemented with high reliability while suppressing the phase interference.
(66) According to exemplary embodiments, the touch sensor pads **150** and the antenna pads **250** may form a single pad row at one end of the sensor device. The single pad row may further include the blocking pad **151** and the guard pad **153**.
(67) The sensing electrodes **110** and **130**, the traces **140** and/or the antenna unit **200** may include silver (Ag), gold (Au), copper (Cu), aluminum (Al), platinum (Pt), palladium (Pd), chromium (Cr), titanium (Ti), tungsten (W), niobium (Nb), tantalum (Ta), vanadium (V), iron (Fe), manganese (Mn), cobalt (Co), nickel (Ni), zinc (Zn), tin (Sn) or an alloy containing at least one of these. These may be used alone or in combination of two or more.
(68) In one embodiment, the sensing electrodes **110** and **130**, the traces **140** and/or the antenna unit **200** may include silver (Ag) or a silver alloy (e.g., a silver-palladium-copper (APC) alloy), or copper (Cu) or a copper alloy (e.g., a copper-calcium (CuCa) alloy) for implementation of low resistance and fine linewidth.
(69) In some embodiments, the sensing electrodes **110** and **130**, the traces **140**, and/or the antenna unit **200** may include a transparent conductive oxide, such as indium tin oxide (ITO), indium zinc oxide (IZO), indium zinc tin oxide (IZTO), or zinc oxide (ZnOx).
(70) In some embodiments, the sensing electrodes **110** and **130**, the traces **140**, and/or the antenna unit **200** may include a lamination structure of a transparent conductive oxide layer and a metal layer, for example, a two-layer structure of a transparent conductive oxide layer-metal layer or a three-layer structure of a transparent conductive oxide layer-metal layer-transparent conductive oxide layer. In this case, the flexible characteristic may be enhanced by the metal layer, and the

signal transmission speed may be improved by reducing the resistance, as well as the corrosion resistance and transparency may be improved by the transparent conductive oxide layer.
(71) The sensing electrodes **110** and **130**, the traces **140** and/or the antenna unit **200** may include a blackening treated part. Accordingly, the reflectivity on the surface of the sensing electrodes **110** and **130**, the traces **140** and/or the antenna unit **200** may be reduced, thereby decreasing pattern visibility due to light reflection.
(72) In one embodiment, the surfaces of the metal layer included in the sensing electrodes **110** and **130**, the traces **140** and/or the antenna unit **200** may be converted into a metal oxide or metal sulfide to form a blackening layer. In one embodiment, blackening layer(s) such as a black material coating layer or a plating layer may be formed on the sensing electrodes **110** and **130**, the traces **140** and/or the antenna unit **200** or the metal layer. The black material or plating layer may include an oxide, sulfide, alloy, or the like containing silicon, carbon, copper, molybdenum, tin, chromium, molybdenum, nickel, cobalt, or at least one of these.
(73) The composition and thickness of the blackening layer may be adjusted considering the reflectivity reduction effect and antenna radiation characteristics.
(74) In one embodiment, the sensing electrodes **110** and **130** and the radiator **210** may include a mesh structure. Accordingly, it is possible to suppress the sensing electrodes **110** and **130** and the radiator **210** from being viewed by a user in the active area AA.
(75) In one embodiment, at least a portion of the transmission line **220** may include a solid structure formed of the above-described metal or alloy. Accordingly, an increase in the resistance between the antenna unit **200** and the antenna pad **250** may be suppressed, and the power supply efficiency may be improved.
(76) The antenna unit **200** is disposed in the active area AA, and may be partially disposed in the peripheral area PA. For example, the transmission line **220** may be disposed in the peripheral area PA. In one embodiment, the transmission line **220** may extend across the active area AA and the peripheral area PA.
(77) In one embodiment, the transmission line **220** may also share the mesh structure. The transmission line **220** may also have a partially solid structure.
(78) In one embodiment, the sensing electrodes **110** and **130** and/or the radiator **210** may include a solid thin film transparent metal structure. Accordingly, the sensing sensitivity and emission performance may be further improved.
(79) The touch sensor pads **150** and the antenna pads **250** may have a solid structure formed of the above-described metal or alloy. Accordingly, the bonding resistance with the circuit board **300** may be reduced.
(80) In exemplary embodiments, the sensing electrodes **110** and **130** and the antenna unit **200** may be disposed in the same layer or at the same level. For example, the sensing electrodes **110** and **130** and the radiator **210** may be formed together in the same layer through the same patterning process to have a mesh structure.
(81) In some embodiments, the antenna unit **200** and the traces **140** may be formed in the same layer or at the same level.
(82) According to exemplary embodiments, the touch sensor pads **150** and the antenna pads **250** may be arranged together in the same layer or at the same level. Accordingly, antenna bonding and touch sensor bonding may be implemented together through one circuit board **300** as described below.
(83) In some embodiments, the antenna unit **200** and the antenna pad **250** may be directly connected with each other in the same layer. Accordingly, antenna signal loss may be suppressed while reducing antenna power supply resistance.
(84) In some embodiments, the sensing electrodes **110** and **130**, the traces **140** and the touch sensor pads **150** may be directly connected with each other in the same layer. This may enhance the touch sensing sensitivity while reducing the sensing channel resistance.

(85) As shown in FIG. **1**, a plurality of antenna units **200** may be arranged in the form of array at one end adjacent to the bonding area BA of the active area AA.
(86) The sensing electrodes **110** and **130** adjacent to the radiator **210** of the antenna unit **200** may have a reduced area compared to the other sensing electrodes **110** and **130**. Accordingly, an arrangement space of the antenna unit **200** may be secured while maintaining the mutual independence of touch sensing and antenna radiation. For example, the sensing electrodes **110** and **130** adjacent to the radiator **210** may include a recess etched so that the radiator **210** is partially inserted.
(87) In some embodiments, the touch sensor pad **150** (e.g., the first touch sensor pad **152**) may be disposed between the adjacent antenna pads **250**. For example, a plurality of antenna pad units may be arranged repeatedly in the row direction with the touch sensor pad **150** interposed therebetween.
(88) In some embodiments, antenna ground pads (not shown) may be further arranged around the antenna pad **250**. Due to the antenna ground pad, mutual independence and reliability of antenna power supply/radiation through the antenna ground pad and touch sensing through the sensing electrodes **110** and **130** may be improved.
(89) In some embodiments, a plurality of touch sensor pads **150** may be disposed between the adjacent antenna pads **250**. For example, a plurality of first touch sensor pads **152** may be disposed between the adjacent antenna pads **250**. In this case, a plurality of first traces **142** may be disposed between the adjacent antenna units **200**.
(90) In some embodiments, one touch sensor pad **150** may be disposed between the adjacent antenna pads **250**. For example, one first touch sensor pad **152** may be disposed between the adjacent antenna pad units. Accordingly, a pad density in the pad row may be increased, while decreasing the number of touch sensor pads **150** sandwiched between the antenna pad units.
(91) FIGS. **3** and **4** are schematic cross-sectional views illustrating a sensor device package according to exemplary embodiments. FIG. **5** is a schematic partially enlarged plan view illustrating the sensor device package according to exemplary embodiments. FIG. **5** is a plan view illustrating an enlarged structure around the bonding area of the sensor device.
(92) FIG. **3** illustrates a connection of the antenna pad **250** with the antenna unit **200** through the circuit board **300**. FIG. **4** illustrates a connection of the touch sensor pad **150** with the sensing electrodes **110** and **130** of the touch sensor through the circuit board **300**. For the convenience of description, the sensor device package is shown by separating in FIGS. **3** and **4**. However, as shown in FIG. **5**, one circuit board **300** and one sensor device may be coupled to implement a single sensor device package.
(93) Referring to FIGS. **3** to **5**, the circuit board **300** may include a core layer **310**, a first conductive layer **320** and a second conductive layer **330**.
(94) The core layer **310** may include a flexible resin such as a polyimide resin, a modified polyimide (MPI), an epoxy resin, a polyester, a cycloolefin polymer (COP), a liquid crystal polymer (LCP) or the like. In a preferred embodiment, the core layer **310** may include the polyimide resin or MPI.
(95) The conductive layers **320** and **330** of the circuit board **300** may include silver (Ag), gold (Au), copper (Cu), aluminum (Al), platinum (Pt), palladium (Pd), chromium (Cr), titanium (Ti), tungsten (W), niobium (Nb), tantalum (Ta), vanadium (V), iron (Fe), manganese (Mn), cobalt (Co), nickel (Ni), tin (Sn), zinc (Zn), molybdenum (Mo), calcium (Ca), or an alloy containing at least one of these metals. In some embodiments, the conductive layers **320** and **330** may include copper or a copper alloy in consideration of signal efficiency and ground efficiency.
(96) In one embodiment, the circuit board **300** may be made from a copper clad laminate (CCL). The circuit board **300** may be provided as a flexible printed circuit board (FPCB).
(97) The core layer **310** may have a first surface **310***a* and a second surface **310***b* which face each other. For example, the first surface **310***a* and the second surface **310***b* may correspond to a lower surface and an upper surface of the core layer **310**, respectively. According to exemplary

embodiments, the first surface **310***a* may correspond to a bonding surface to the sensor device, and the second surface **310***b* may correspond to a mounting surface of a driving circuit/driving circuit connection structure.

(98) The first conductive layer **320** and the second conductive layer **330** may be formed on the first surface **310***a* and the second surface **310***b* of the core layer **310**, respectively.

(99) The first conductive layer **320** may include a first antenna signal line **322** and a first touch sensor signal line **325**. The second conductive layer **330** may include a second antenna signal line **332** and a second touch sensor signal line **335**.

(100) The first antenna signal line **322** and the first touch sensor signal line **325** may be distributed together at the same level on the first surface **310***a* of the core layer **310**. The second antenna signal line **332** and the second touch sensor signal line **335** may be distributed together at the same level on the second surface **310***b* of the core layer **310**.

(101) As illustrated in FIG. **3**, the first antenna signal line **322** may be electrically connected to the antenna pad **250** by a bonding intermediary structure **180**. According to exemplary embodiments, the bonding intermediary structure **180** may include an anisotropic conductive film (ACF).

(102) As illustrated in FIG. **1**, the plurality of antenna units **200** may be arranged in an array form. In this case, a plurality of first antenna signal lines **322** may be individually bonded or connected to the antenna pads **250** to correspond to each of the antenna units.

(103) The second antenna signal line **332** may be electrically connected to the first antenna signal line **322** on the second surface **310***b* of the core layer **310**. For example, a plurality of second antenna signal lines **332** may be arranged on the second surface **310***b* of the core layer **310** to correspond to each of the plurality of first antenna signal lines **322**.

(104) According to exemplary embodiments, the first antenna signal line **322** and the second antenna signal line **332** may be electrically connected to each other through an antenna via structure **324**. The antenna via structure **324** may penetrate the core layer **310** to connect the first antenna signal line **322** and the second antenna signal line **332**.

(105) For example, the antenna via structure **324** may be formed by forming an antenna via hole to penetrate the core layer **310**, and filling the antenna via hole through plating (e.g., copper plating) process. The antenna via structure **324** may be formed as a conductor substantially integral with the first antenna signal line **322** and the second antenna signal line **332**.

(106) The second antenna signal line **332** may be electrically connected with an antenna driving circuit/antenna connection structure **350**. In one embodiment, the antenna driving circuit/antenna connection structure **350** may include an antenna connector. The antenna connector may be, for example, a board-to-board (B2B) connector. In this case, the circuit board **300** may be coupled to a chip mounting substrate through the antenna connector.

(107) For example, the antenna driving circuit/antenna connection structure **350** may include an antenna driving integrated circuit (IC) chip. In this case, an antenna driving IC chip **350***b* (see FIG. **9**) may be directly mounted on the circuit board **300** through the antenna conductive intermediary structure **340**. The antenna conductive intermediary structure **340** may include a solder, a conductive ball, a conductive wire and the like.

(108) As illustrated in FIG. **4**, the first touch sensor signal line **325** may be bonded to the touch sensor pad **150** to be electrically connected thereto by the above-described bonding intermediary structure **180**.

(109) A plurality of first touch sensor signal lines **325** may be individually bonded or connected to the touch sensor pads **150** to correspond to each of the traces **140**.

(110) The second touch sensor signal line **335** may be electrically connected with the first touch sensor signal line **325** on the second surface **310***b* of the core layer **310**. For example, a plurality of second touch sensor signal lines **335** may be arranged on the second surface **310***b* of the core layer **310** to correspond to each of the first touch sensor signal lines **325**.

(111) According to exemplary embodiments, the first touch sensor signal line **325** and the second

touch sensor signal line **335** may be electrically connected to each other through a touch sensor via structure **327**. The touch sensor via structure **327** may penetrate the core layer **310** to connect the first touch sensor signal line **325** and the second touch sensor signal line **335**.

(112) For example, the touch sensor via structure **327** may be formed by forming a touch sensor via hole to penetrate the core layer **310**, and filling the touch sensor via hole through the plating (e.g., copper plating) process. The touch sensor via structure **327** may be formed as a conductor substantially integral with the first touch sensor signal line **325** and the second touch sensor signal line **335**.

(113) The second touch sensor signal line **335** may be electrically connected with the touch sensor driving circuit/touch sensor connection structure **370**. In one embodiment, the touch sensor drive circuit/touch sensor connection structure **370** may include a touch sensor connector. The touch sensor connector may be, for example, a board-to-board (B2B) connector. In this case, the circuit board **300** may be coupled to the chip mounting board through the touch sensor connector.

(114) For example, the touch sensor driving circuit/touch sensor connection structure **370** may include a touch sensor driving integrated circuit (IC) chip. In this case, the touch sensor driving IC chip **370***b* may be directly mounted on the circuit board **300** through a touch sensor conductive intermediary structure **360**. The touch sensor conductive intermediary structure **360** may include a solder, a conductive ball, a conductive wire and the like.

(115) According to the above-described exemplary embodiments, the touch sensor pads **150** and the antenna pads **250** may be arranged together in the bonding area BA of the sensor device, such that power supply/signal transmission of the antenna and the touch sensor may be performed together through one circuit board **300**.

(116) The circuit board **300** is formed as a multilayer conductive structure including the first conductive layer **320** and the second conductive layer **330**, and the first conductive layer **320** and the second conductive layer **330** may be designed to include an antenna signal line and a touch signal line, respectively. Through the antenna via structure **324** and the touch sensor via structure **327** which connect the first conductive layer **320** and the second conductive layer **330** with each other, the power supply reliability and power supply efficiency to the antenna pad **250** and the touch sensor pad **150** may be improved using the single circuit board **300**.

(117) In some embodiments, a reference potential or ground potential may be applied to the blocking pad **151** and/or the guard pad **153**. For example, a touch drive signal potential may be applied to the touch sensor pads **152** and **154** through the touch sensor drive IC chip **370***b* (see FIG. **9**), and the reference potential or ground potential may be applied to the blocking pad **151** and/or the guard pad **153**.

(118) The reference potential or ground potential may be lower than each of a potential of the touch drive signal to the touch sensor pad **150** and a power supply potential to the signal pad **252**. In one embodiment, the reference potential or ground potential may be substantially 0 V.

(119) In some embodiments, the touch sensor signal lines **325** and **335** may further include signal lines which are connected to the blocking pad **151** and/or the guard pad **153** to apply the reference potential or ground potential thereto.

(120) In some embodiments, the reference potential or ground potential may be applied to the antenna ground pad (not shown).

(121) Mutual independence of the antenna power supply/driving signals and the touch sensing signals may be enhanced through the above-described reference potential/ground potential.

(122) FIGS. **6** and **7** are schematic cross-sectional views illustrating the sensor device package according to exemplary embodiments. Configurations and structures that are substantially the same as or similar to those described with reference to FIGS. **1** to **5** will not be described in detail.

(123) Referring to FIGS. **6** and **7**, the circuit board **300** may further include a mid-ground layer **390** disposed between the first conductive layer **320** and the second conductive layer **330**. In this case, the circuit board **300** may have a conductive layer structure of three or more layers.

(124) The core layer **310** may be provided in the form of a plurality of interlayer insulation layers. According to exemplary embodiments, a first interlayer insulation layer **312** may be disposed between the first conductive layer **320** and the mid-ground layer **390**, and a second interlayer insulation layer **314** may be disposed between the mid-ground layer **390** and the second conductive layer **330**.
(125) The mid-ground layer **390** may be electrically and physically separated from the antenna via structure **324** and the touch sensor via structure **327**. In one embodiment, the mid-ground layer **390** may include holes through which the antenna via structure **324** and the touch sensor via structure **327** penetrate, respectively.
(126) In one embodiment, a blocking insulation layer **395** may be formed between the mid-ground layer **390** and the via structures **324** and **327**. Accordingly, insulating separation between the mid-ground layer **390** and the via structures **324** and **327** may be stably maintained.
(127) The independence and efficiency of each of the antenna signal transmission and the touch sensor signal transmission through the first conductive layer **320** and the second conductive layer **330** may be increased by the mid-ground layer **390**, and the overall signal/power loss may be reduced.
(128) FIGS. **8** and **9** are a schematic plan view and a cross-sectional view illustrating an image display device according to exemplary embodiments, respectively.
(129) FIG. **8** illustrates a front portion or a window surface of an image display device **400** manufactured in the form of, for example, a smart phone. The front surface of the image display device **400** may include a display area (DA) **430** and a non-display area (NDA) **440**. The non-display area **440** may correspond to, for example, a light-shielding part or a bezel part of the image display device **400**.
(130) Referring to FIGS. **8** and **9**, the image display device **400** may include a display panel **410** and the above-described sensor device disposed on the display panel **410**.
(131) The sensor device according to the above-described exemplary embodiments may be disposed toward the front portion of the image display device **400**, and for example, may be disposed on the display panel. Accordingly, the antenna units **200** included in the sensor device may be provided as an antenna-on-display (AOD) antenna.
(132) In some embodiments, the sensor device may be attached to the display panel in the form of a film. In some embodiments, the sensor device may be disposed across the display area **430** and the non-display area **440** of the image display device **400**.
(133) In some embodiments, the active area AA of the sensor device may be overlapped with the display area **430**. In one embodiment, the sensing electrodes **110** and **130** and/or the radiator **210** may be at least partially overlapped with the display area **430**.
(134) In some embodiments, the peripheral area PA of the sensor device may be overlapped with the non-display area **440**. The traces **140** and/or the signal pads **230** of the antenna unit **200** may be at least partially overlapped with the non-display area **440**. For example, a portion having a solid structure of the sensor device may be overlapped with the non-display area **440**.
(135) The sensor device may be powered or driven through the circuit board **300**. As described with reference to FIGS. **3** and **4**, the circuit board **300** is electrically connected with the antenna driving circuit/antenna connection structure **350**. The antenna driving circuit/antenna connection structure **350** may include an antenna connector **350***a* and an antenna driving IC chip **350***b*. The touch sensor driving circuit/touch sensor connection structure **370** may include a touch sensor connector **370***a* and a touch sensor driving IC chip **370***b*.
(136) For the convenience of illustration, the antenna connector **350***a* and the touch sensor connector **370***a* are shown as one connector in FIG. **9**. However, as described above, the antenna connector **350***a* and the touch sensor connector **370***a* may be mounted as independent, separate members on the circuit board **300**. The antenna driving IC chip **350***b* and the touch sensor driving IC chip **370***b* may also be mounted as independent, separate chips on a chip mounting board **450**.

(137) The circuit board **300** may be bent to the back portion of the image display device **400** to be connected with the driving IC chips **350***b* and **370***b* mounted on the chip mounting board **450**.
(138) For example, the antenna unit **200** and the circuit board **300** may be electrically connected with the antenna driving IC chip **350***b* through the antenna connector **350***a*. The sensing electrodes **110** and **130** of the touch sensor and the circuit board **300** may be electrically connected with the touch sensor driving IC chip **370***b* through the touch sensor connector **370***a*.
(139) The chip mounting board **450** may be a rigid printed circuit board, and for example, may be a main board of the image display device **400**.
(140) As described with reference to FIGS. **3** and **4**, the antenna driving IC chip **350***b* and the touch sensor driving IC chip **370***b* may be directly mounted on the circuit board **300**.
(141) According to exemplary embodiments, the display panel **410** may further include an optical layer **420** disposed thereon. For example, the optical layer **420** may be a polarizing layer including a polarizer or a polarizing plate. In some embodiments, the sensor device may be disposed on the optical layer **420**.

Claims

1. A sensor device package comprising: a sensor device including touch sensing electrodes and an antenna unit; and a circuit board bonded to the sensor device, the circuit board comprising: a core layer comprising a first surface and a second surface which face each other; a first conductive layer comprising first touch sensor signal lines and a first antenna signal line which are distributed at the same level on the first surface of the core layer; a second conductive layer comprising second touch sensor signal lines and a second antenna signal line which are distributed at the same level on the second surface of the core layer; an antenna via structure which penetrates the core layer to connect the first antenna signal line and the second antenna signal line with each other; and a touch sensor via structure penetrating the core layer to connect each of the first touch sensor signal lines and the second touch sensor signal lines with each other.
2. The sensor device package according to claim 1, wherein the sensor device further comprises touch sensor pads electrically connected with the touch sensing electrodes and an antenna pad electrically connected with the antenna unit; the first touch sensor signal lines of the circuit board are bonded to the touch sensor pads respectively; and the first antenna signal line of the circuit board is bonded to the antenna pad.
3. The sensor device package according to claim 2, wherein the touch sensor pads and the antenna pad are arranged at one end of the sensor device to form a single pad row.
4. The sensor device package according to claim 3, wherein the sensor device further comprises a blocking pad which is included in the single pad row and inserted between adjacent touch sensor pads of the touch sensor pads, or a guard pad disposed at an end of the single pad row.
5. The sensor device package according to claim 4, wherein the sensor device further comprises: traces which connect the touch sensing electrodes and the touch sensor pads with each other; and a blocking line which extends from the blocking pad between adjacent traces of the traces.
6. The sensor device package according to claim 4, wherein the sensor device further comprises a guard line of a loop shape, which extends from the guard pad and surrounds the touch sensing electrodes.
7. The sensor device package according to claim 3, further comprising one anisotropic conductive film which bonds the single pad row and the first conductive layer of the circuit board.
8. The sensor device package according to claim 2, wherein the antenna unit comprises a radiator and a transmission line which extends from the radiator to be connected to the antenna pad.
9. The sensor device package according to claim 8, wherein a plurality of antenna units are arranged in a row direction, and the antenna pads are independently connected to each of the antenna units.

10. The sensor device package according to claim 9, wherein the transmission lines connected to each of the plurality of antenna units have a bent line shape and have the same length.
11. The sensor device package according to claim 9, wherein at least one touch sensor pad of the touch sensor pads is arranged between the antenna pads connected to adjacent antenna units of the antenna units.
12. The sensor device package according to claim 1, further comprising: a touch sensor driving circuit/touch sensor connection structure electrically connected with the second touch sensor signal lines on the second surface of the core layer; and an antenna driving circuit/antenna connection structure electrically connected with the second antenna signal line on the second surface of the core layer.
13. The sensor device package according to claim 12, wherein the antenna driving circuit/antenna connection structure comprises an antenna driving integrated circuit chip or an antenna connector, and the touch sensor driving circuit/touch sensor connection structure comprises a touch sensor driving integrated circuit chip or a touch sensor connector.
14. The sensor device package according to claim 1, wherein the circuit board further comprises a mid-ground layer disposed between the first conductive layer and the second conductive layer in the core layer.
15. The sensor device package according to claim 14, wherein the antenna via structure and the touch sensor via structure are electrically separated from the mid-ground layer and penetrate the mid-ground layer.
16. The sensor device package according to claim 1, wherein the sensor device further comprises a substrate layer, wherein the touch sensing electrodes and the antenna unit are arranged together at the same level on the upper surface of the substrate layer.
17. The sensor device package according to claim 16, wherein the substrate layer comprises an active area where the touch sensing electrodes are distributed and a peripheral area surrounding the active area, wherein the antenna unit is partially disposed on the active area.
18. An image display device comprising: a display panel; and the sensor device package according to claim 1.